INCH-POUND

MIL-M-43576A
7 April 1989
SUPERSEDING
MIL-M-43576
29 March 1968

MILITARY SPECIFICATION

MANTLE, ILLUMINATING (GASOLINE LANTERN)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of knitted, soft-type mantle for use in a gasoline pressure lantern (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-636 - Box, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 6260

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MILITARY

MIL-L-1594 - Lantern, Gasoline, Leaded or Nonleaded Fuel, with Case
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members For Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Packaging Commercial

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Materials and components. The materials and components used shall conform to the applicable referenced specifications, standards, and requirements specified herein. Where materials are not definitely specified, they shall be of the quality normally used for the purpose in commercial practice provided the completed items comply with all provisions of this specification. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3 Design and construction. The mantle (see figure 1) shall be made from a chemically treated seamless tubing knitted of rayon yarn. The bottom or closed end of the mantle shall be formed by gathering or pleating the tube and sewing with chemically treated cotton thread. The resulting bag shall then be turned so that the stub end is on the inside. The top section of the mantle shall be treated with a chemical solution to harden and increase the strength and abrasion resistance of this section of the mantle after ignition.. Approximately half of this treated band shall be turned in, to form a double thickness. An asbestos drawstring shall be threaded around and through the top or head hardened section of the mantle for fastening the mantle holder of the lantern.

3.3.1 Seamless knitted tubing. The seamless tubing used in the manufacture of the mantle shall be knit from 150 denier, 42 filament, bright rayon yarn using a "duplex weave" or stitch.

3.3.2 Gathering or pleating thread. The thread used to close the bottom of the tube of the mantle shall be a white chemically treated cotton thread.

3.3.3 Head hardening treatment. The top 1-1/4 inches (+3/16, -1/8 inch) of the mantle shall be treated with a chemical solution to increase the after ignition strength of this section of the mantle. The hardening solution shall contain a dye which will show the extent of treatment.

3.3.4 Asbestos drawstring. The drawstring shall be a 4 ply, 35 cut asbestos thread, 7 inches (+', -1/2 inch) in length.

3.4 Size. The mantle, as received and prior to ignition, shall conform to the dimensional requirements specified in figure 1. The dimensions are based on the flat configuration of the mantle.

3.5 Performance. After ignition, the mantle shall assume an overall symmetrical shape and retain this shape throughout the test, when tested as specified in 4.4.4. At the conclusion of the test, the mantle shall measure at least 1-1/2 inches in length. Not more than one in ten mantles tested shall show a hole greater than 1/4 inch in diameter resulting from broken threads.

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3.6 Light output. The mantle shall have a light output of not less than 120 candle power, when tested as specified in 4.4.4.

3.7 Weight of ash in mantle. The weight of ash in the mantle (with drawstring removed) shall be not less than 0.320 grams, when tested as specified in 4.4.4.

3.8 Workmanship. The mantle shall be free of tears, rips, and holes that may affect the performance of the mantle.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3 and tested for the characteristics specified in table I. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the components and materials conform to the requirements specified in 3.3.1 through 3.3.4.

4.4.2 End item visual examination. The end items shall be examined for the defects listed below. The lot size shall be expressed in units of mantles. The sample unit shall be one mantle. The inspection level shall be S-4 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Mantle	Bottom of mantle unraveled.
	Stub end on the outside.
	Top of the mantle not dyed.
	Treated band not doubled over.
	Drawstring missing; not threaded around and through the top.
	Drawstring less than 6-1/2 inches.
	Drawstring more than 8 inches.
	Any tear, rip, or hole that may affect the performance of the mantle.

4.4.3 End item dimensional examination. The end items shall be examined for the defects listed below. The lot size shall be expressed in units of mantles. The sample unit shall be one mantle. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Mantle	
Width	Less than 2-1/8 inches. More than 2-3/8 inches.
Length (head hardened section turned in)	Less than 2-5/8 inches. More than 3-1/8 inches.
Head hardened area (full length, see 3.3.3)	Less than 1-1/8 inches. More than 1-7/16 inches.
Head hardened area (doubled over)	Less than 5/8 inch. More than 7/8 inch.

4.4.4 End item testing. The end item shall be tested for the characteristics listed in table I. The methods of testing specified in FED-STD-191 wherever applicable and listed in table I shall be followed. The lot size shall be expressed in units of mantles and the sample unit shall be twelve mantles. Any test failure shall be cause for rejection of the lot.

Lot size (mantles)	No. of sample units
4800 or less	12
4801 up to and including 132,000	18
132,001 or more	30

TABLE I. End item tests

Characteristic	Requirement paragraph	Test method	Number of determinations per sample unit	Results reported as
Performance	3.5	4.5.3.2	10	Pass or fail
Light output	3.6	4.5.3.3	1	Pass or fail
Total weight of ash	3.7	Method 2611	2	Average of two determinations to nearest 0.001 grams

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4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Markings, (exterior and interior)	Omitted, incorrect, incomplete, illegible; of improper size, location, sequence, or method of application.
Materials	Any nonconforming component; component missing or damaged.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps; loose or inadequate sealing, strapping, or stapling.
	Envelope not fabricated from material specified.
	Envelope not securely closed.
	Bulged or distorted container.
Contents	Number of unit packs per intermediate pack or packs per shipping container is more or less than specified. 1/

1/ No more than five intermediate packs from a shipping container shall be taken for the unit pack examination.

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified

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Examine	Defect
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 METHODS OF INSPECTION

4.5.1 Performance and light output tests.

4.5.1.1 Equipment.

4.5.1.1.1 Lantern. A single-mantle gasoline lantern conforming to MIL-L-1594, with a pressure guage installed, shall be used. Nonleaded gasoline shall be used as fuel during the tests.

4.5.1.1.2 Light meter. A foot-candle meter equipped with Viscor and cosine correction filters shall be used (Weston Model 614 or equal). The meter shall be fitted with a 4-inch diameter tube, lined with black felt, in front of the cell. The length of the tube, when placed against the lantern globe, shall space the face of the cell a distance of 17 inches (1.414 feet) from the center of the mantle.

4.5.1.2 Performance test. Install one of the test mantles in the lantern and light in accordance with the instruction specified in MIL-L-1594, with an initial air pressure of 20 to 22 pounds per square inch (psi). Operate the lantern for 30 minutes, turn off, and allow to cool for 30 minutes. Repeat this operation for a total of five times. During the test, measure the length of the mantle. Use this mantle for light output measurement. Install and ignite the nine remaining mantles, igniting each mantle only once. Observe broken threads and measure the size of any hole.

4.5.1.3 Light output. Using the multi-ignited mantle as tested in 4.5.1.2, light the lantern and operate in accordance with instructions above, maintaining an air pressure of 20 to 22 psi. Place the ignited lantern in a normally illuminated room, excluding any other intense light source, natural or artificial, and measure the light output of the mantle using the foot-candle meter fitted with the tube specified in 4.5.1.1.2. Calculate the light output in candlepower as follows:

$$\text{Candlepower} = \text{foot-candles} \times D^2$$

where:

D = distance in feet from the center of mantle to the sensing cell of meter.

(Note: When D = 1.414 feet, then Candlepower = Foot-candles X 2)

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit pack. Each mantle shall be unit packed in a close-fitting coin-type envelope fabricated of kraft or glassine paper, or polyethylene film bag. The envelope flap shall be closed and sealed. The polyethylene bag closure shall be heat sealed.

5.1.1.2 Intermediate pack. Six mantles, unit packed as specified in 5.1.1.1, shall be packed in a snug-fitting intermediate paperboard box conforming to variety plain, style II, type A, class a of PPP-B-566. Each box shall be closed in accordance with the appendix of PPP-B-566.

5.1.2 Commercial. Mantles shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A packing. One thousand seven hundred and twenty-eight mantles, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A intermediate packs shall be packed flat 6 in length, 4 in width, and 12 in depth within a shipping container. Inside dimensions of each shipping container shall approximate 26 inches in length, 13 inches in width, and 9-3/4 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. One thousand seven hundred and twenty-eight mantles, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A intermediate packs shall be packed flat 6 in length, 4 in width, and 12 in depth within a shipping container. Inside dimensions of each shipping container shall approximate 26 inches in length, 13 inches in width, and 9-3/4 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

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5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636.

5.2.3 Commercial packing. Mantles, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), mantles, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 3 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Precautionary marking. Each unit pack shall have the following warning. Precaution: Wash hands after handling the mantle. The mantle drawstring contains asbestos fiber which can be harmful if inhaled.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The mantle is intended for use in a single-mantle gasoline lantern which conforms to the requirements of MIL-L-1594; Lantern, Gasoline, Leaded or Nonleaded Fuel, with Case.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
e. Type and class of unit load required (see 5.2.1).
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
g. When palletization is required (see 5.3).

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Burn
Fixture
Gas
Lamp
Light

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Air Force - 99

Review activities:

Army - MD
Air Force - 82
DLA - GS

User activities:

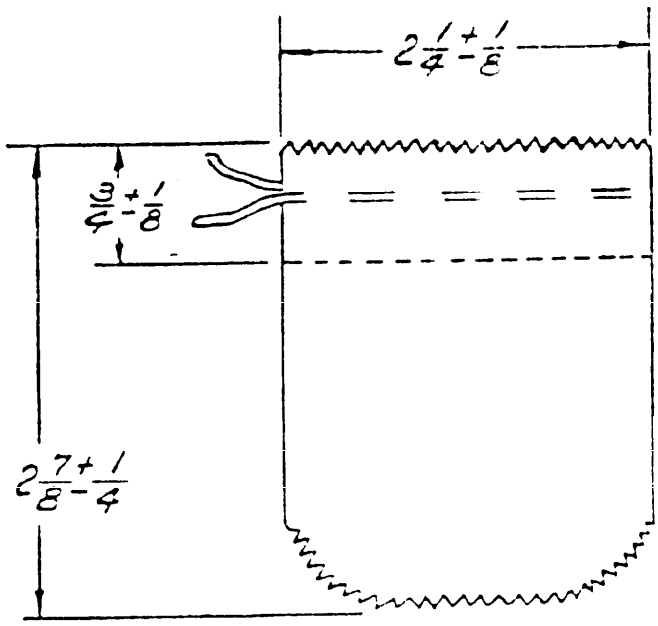
Army - ME
Navy - CG, MC

Preparing activity:

Army - GL

(Project 6260-1001)

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WITH HEAD HARDENED
SECTION TURNED IN

FIG. 1 MANTLE, ILLUMINATING (GASOLINE LANTERN)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-43576A	Mantle, Illuminating (Gasoline Lantern)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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